Preference ≠ Tattooability: Auditing Reward Models for AI Tattoo Generation

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Abstract

AI tattoo generation is usually ranked with the same reward models that rank general text-to-image outputs. Those models predict which image a crowd annotator prefers. A tattoo that will be worn also has to survive a needle, a gap that heals shut, and ink that spreads. We audited that gap with deterministic, physically scaled probes and with CLIPScore, PickScore, HPSv2.1 and ImageReward, on tattoo-subject pairs from HPDv2 and ImageRewardDB, on controlled defects, and on a print-scale ink-spread stress test. On HPDv2 tattoo-subject pairs, a-priori probes agree with human preference below chance, and that contrast remains after a Holm correction across scorers. On ImageRewardDB (9 prompts) and Pick-a-Pic (11 prompts) the same probes are not significant after Holm. A prompt-held-out logistic fusion on HPDv2 rises above chance only by learning inverted signs. Reward models track preference, and they also flag many injected defects, but HPSv2.1 was trained on HPDv2 and ImageReward on ImageRewardDB, so those agreements are in-distribution. The probes are interpretable and tied to millimetre assumptions. The rewards are not. Nothing here ranks a commercial AI tattoo generator.

Keywords: AI tattoo generation; AI tattoo generator; human preference; reward models; tattooability.

1 Introduction

Text-to-image systems are aligned and ranked with human preference [1–7]. The question those benchmarks answer is perceptual: which image do people prefer for this prompt? AI tattoo generation asks a second question. If the image is printed on skin at a stated size, do the lines stay wider than a needle, do the gaps stay open after a small amount of ink spread, and does the shading stay a deliberate tone rather than a muddy mid-grey? Those properties are not what preference annotators were asked to judge.

Public preference sets already mix flash-sheet prompts with portraits that merely wear tattoos. A reward can rank the portraits and still say nothing about whether a design survives a needle. Tattoo vision work covers recognition, retrieval and removal [8–11], not this mismatch.

We measured the mismatch with a small set of CPU probes. Each probe rescales the image so that the longest side is a chosen print size, then reports a length, a fraction or a density in millimetres. Signs were fixed before looking at preference accuracy: thicker minimum lines are better; sub-needle lines, open stroke ends, tight gaps, mid-tones, uneven borders, edge clutter and specks are worse. One further probe, edge-transition width, was added after the first controlled-degradation run. It is reported as post hoc. An unweighted composite of the signed probes is reported and then set aside. Per-dimension numbers are the ones we interpret.

Contributions.

1. A documented probe suite (Section 3) with needle width 0.30 mm, gap width 0.50 mm, and analysis resolution 10 px/mm, evaluated at print sizes 3, 5, 10 and 15 cm.

2. A prompt-clustered comparison of each probe, and of CLIPScore, PickScore, HPSv2.1 and ImageReward, with human winners on tattoo-subject pairs (Section 5). The resampling unit is the prompt. On HPDv2 the pre-registered probes sit below chance, and 11 of 13 scorers remain significant after Holm. On ImageRewardDB none of the 13 scorers is Holm-significant. On Pick-a-Pic the only Holm-significant scorer is PickScore (1 of 13); the probes are not. Those two sets have 9 and 11 prompts. A prompt-held-out logistic fusion on HPDv2 reaches 0.708 only with inverted coefficients.

3. A controlled-defect check on 120 public designs: single probes detect the defects they were built for, and the reward models detect many defects as well. HPSv2.1 is the strongest of the four rewards on that check.

4. A scale-by-spread stress test under an isotropic Gaussian blur, with no additive pixel noise in the default grid. It is not a model of healed skin. With that default, skeleton F_1 at zero spread is exact, and it falls as the spread is widened.

The author operates a commercial AI tattoo generator. That product was not scored, and none of its outputs are in the datasets (Section 8).

2 Related work

Preference models for text-to-image systems. Human Preference Score and HPSv2 fit a scorer on crowd choices over generated images, and HPDv2 is the dataset released with that line of work [2, 3]. ImageReward learns a scalar from rated text–image pairs in ImageRewardDB [4]. Pick-a-Pic collects web user choices and trains PickScore [5]. CLIPScore uses CLIP cosine similarity as a reference-free match between a caption and an image [12, 13]. Direct preference optimisation and policy gradients have been used to fine-tune diffusion models against such rewards [6, 7]. Latent diffusion is the generator family that produced much of this training imagery [1], and LAION-scale data sit underneath the CLIP encoders [14]. Compositional benchmarks, holistic grids and localised human marks still score a picture, not a mark at a print size [15–18].

Line drawings. Suggestive contours and the "where do people draw lines" studies treat line selection as a perceptual decision: which luminance ridges a viewer would mark, not which strokes a needle can hold [19, 20]. Simplification then asks which of those strokes can be dropped while the depiction remains recognisable [21]. Vectorisation fidelity asks whether the recovered curves still lie on the drawing [22]. Cleanup and virtual-sketching benchmarks score how close a rough stroke becomes to a clean vector, usually in pixels on a screen [23, 24]. Those notions are the right family for tattooability, and they stop at the picture. A tattoo line is selected under a millimetre budget: width against needle width w_n, separation against gap g, and a blur σ that models ink spread after the mark is printed at size s. Our probes are that budget, measured on a physically rescaled raster. They are not a line-selection model, a simplification objective, or a vectorisation F_1 against an artist stroke.

Tattoo images. NIST's Tatt-BP note is a collection protocol for tattoo photographs used in recognition [8]. TattTRN reconstructs templates for retrieval [9]. GenInk synthesises tattoo images for biometric experiments [10]. Synthetic facial tattoos have been used to test recognition under ink [11]. We did not find a published audit of whether preference rewards agree with physical tattooability probes. That absence is a search result, not a proof (Appendix B).

Classical image operators. The probes call Otsu's threshold, Canny edges and Zhang–Suen thinning [25–27]. Structural similarity is the usual reference image-quality score [28]. We do not use it: preference pairs are different images, not a distorted copy of one reference.

3 Method

Notation. Print size s is the longest side in centimetres. Ink spread σ is an isotropic Gaussian standard deviation in millimetres. Needle width is w_n=0.30 mm and the tight-gap width is g=0.50 mm. Pairwise accuracy is A. The effect is $A - 0.5$, and h is Cohen's h of A against 0.5. Skeleton agreement is F_1 at a 0.2 mm tolerance. The same symbols are used in the text, the tables and the captions.

3.1 Physical rescale

Every probe reads an RGB image, converts it to grey, and rescales it so the longest side spans the chosen print size s at 10 px/mm. Downscaling uses area interpolation. Upscaling uses cubic interpolation. Sizes 3, 5, 10 and 15 cm are the grid. Preference tables use the 5 cm probes. The degradation table also records 10 cm probes in the raw file; the paper reports the 5 cm slice.

Ink is the set of pixels darker than an Otsu threshold [25]. If the grey standard deviation is at most 1, the threshold is fixed at mid-grey. Line width is twice the distance transform on the ink mask, read on the Zhang–Suen skeleton and converted to millimetres [27].

3.2 Pre-registered probes

Signs were fixed before any preference accuracy was computed. +1 means a larger value is treated as more tattooable. `line_w_p10_mm` (+1) is the 10th percentile of skeleton line width. `frac_sub_needle` (−1) is the fraction of skeleton pixels thinner than w_n. `open_ends_per_cm` (−1) counts skeleton endpoints per centimetre of skeleton. `frac_tight_gap` (−1) is the fraction of the background skeleton inside the ink box that is narrower than g. `midtone_frac` (−1) is the fraction of pixels strictly between grey 60 and grey 200. `border_std` (−1) is the standard deviation of a 5 px border. `edge_density_per_mm2` (−1) counts Canny edge pixels per square millimetre, with hysteresis 50 and 150 [26]. `specks_per_cm2` (−1) counts connected ink components smaller than a w_n square, per square centimetre. An unweighted composite, the mean of signed robust z-scores, was defined in advance and is deprecated. Appendix A keeps one sentence on it. Per-dimension accuracies in Table 2 are the probe result we interpret. Edge transition (below) is excluded from the pre-registered feature list and included only in the "all probes" fusion.

3.3 Post-hoc edge transition

`edge_transition_mm` (sign −1) estimates how wide the ink-to-paper ramp is. On Canny pixels we take the median Sobel magnitude and divide the median paper-minus-ink contrast by that magnitude, then convert pixels to millimetres. A soft or blurred edge scores higher. This probe was added after the first degradation run showed that blur and detail loss were invisible to the pre-registered set. Section B records that order. We do not treat its preference accuracy as a confirmatory test.

3.4 Scores we compare against

CLIPScore, PickScore, HPSv2.1 and ImageReward are used as released scorers [3–5, 12]. Higher is treated as the model's preferred image. We do not retrain them. HPSv2.1 was trained on HPDv2, and ImageReward was trained on ImageRewardDB. Accuracy of those two models on their own training distribution is leakage, not an external test. PickScore was trained on Pick-a-Pic v1, so its accuracy on our Pick-a-Pic v2 tattoo-subject pairs is also leaky. HPSv2.1 and ImageReward were not trained on Pick-a-Pic. On HPDv2, every reward score covers the same tattoo-subject pairs as the probes.

Pairwise accuracy A gives the human winner one point when its score is strictly higher, and half a point on a tie. Intervals are 95% prompt-cluster bootstrap percentiles from 2000 resamples with seed 0. Each resample draws prompts with replacement and keeps every pair inside a drawn prompt, so the unit is the prompt. A two-sided bootstrap p is then Holm-adjusted within each dataset across all scorers in Table 2. A cell that sits on the smallest resolvable p is reported as $p_{\text{Holm}} \leq 0.006$. The fusion model is L2 logistic regression on the difference of standardised features, with GroupKFold by prompt at k=5 and seed 0. Within each fold the winner side is randomly flipped so the label is balanced. Held-out accuracy counts a pair as correct when the predicted probability that the original orientation is the human order exceeds one half. Coefficients quoted in the text are the signs of a final model fit on all pairs. Their magnitudes are in the released coefficient file, not restated as new claims.

3.5 Controlled defects

On 120 designs we apply six operators at two strengths, always from the clean grey image. Thinning uses a dilation of the grey image with a 3 or 5 pixel square kernel (dark ink shrinks). Thickening uses the matching erosion. Haze blends the image toward a flat grey at weights 0.30 and 0.50 and adds a small Gaussian pixel noise. Speckle sets a random fraction 0.003 or 0.01 of pixels to black. Blur is a Gaussian with σ of 1.5 or 3.0 pixels on the original raster, before physical rescale. Detail loss downsamples by 4 or by 8 with area interpolation and nearest-neighbour upsample. Detection is the fraction of designs on which the clean image outscores the degraded image under the same rule as pairwise accuracy. Reward models were run on a subset of 30 designs. The table uses that subset for every column so the cells match.

3.6 Ink-spread stress test

The spread model rescales to print size s and applies an isotropic Gaussian of spread σ. The default grid adds no pixel noise. An earlier setting added grey-level noise with pixel standard deviation 4 and made skeleton F_1 fall with print size even at zero spread; that decline was an artifact, and those four noise-on values are reported only as a contrast. The grid is sizes 3, 5, 10 and 15 cm and spreads from 0 to 0.60 mm in steps of 0.05 mm. Skeleton F_1 compares the clean ink skeleton with the spread ink skeleton. The reference and the simulation use the same Otsu threshold. A skeleton pixel matches if it lies within 0.2 mm of the other skeleton. Gap survival is the fraction of background pixels inside the clean ink box that remain background after the spread. This is a stress test of the probes' scale assumptions. It is not calibrated to photographs of healed tattoos.

4 Data

4.1 Preference pairs

HPDv2 training pairs whose prompt text matches "tattoo" contribute 2,010 pairs, 327 prompts and 1,340 images. ImageRewardDB training and validation rows with a tattoo prompt contribute 370 rows. 47 of those image files are empty in the upstream archive, leaving 323 usable images. Pick-a-Pic v2 rows from a public parquet mirror contribute 2,608 tattoo rows, of which 1,687 are labelled and show two different images, across 185 prompts. The original Hugging Face dataset card has been removed. Mirrors state an MIT licence. We could not verify that statement, so the mirror is used only as local metadata and local images, and we do not redistribute either. Keyword selection keeps 286 labelled pairs over 23 prompts. Images for those rows, 332 files, were read by row-group ranges from the mirror. CLIP then keeps only design and on-skin pairs.

Table 1: Corpora in this audit. Tattoo-subject images are design or on-skin. HPDv2 and ImageRewardDB ambiguous files (106 images) were relabelled by one non-blinded rater. Pick-a-Pic v2 images were fetched and then CLIP-filtered; only design and on-skin pairs were scored. Pair counts are in Table 2.

	HPDv2	ImageRewardDB	Pick-a-Pic v2
Prompts with "tattoo"	327	—	185
Images or rows	1,340	323 / 370	2,608
Images fetched	—	—	332
Empty files dropped	—	47	—
Design, before review	334	48	116
Ambiguous, before review	99	7	216
Design, after review	377	48	—
On skin, after review	90	12	—
Incidental, after review	682	240	—
Other, after review	191	23	—
Keyword pairs / prompts	—	—	286 / 23
Tattoo-subject prompts	125	9	11

4.2 Tattoo-subject filter

Keyword rules mark an explicit design request or an incidental "person with tattoos" prompt. Otherwise, and when the keyword and the image disagree, we use CLIP ViT-B/32 zero-shot among four captions: a design on paper, a close-up of a tattoo on skin, a picture of a person who has tattoos, and something else [13]. The automatic labels are design, on-skin, incidental, other, or ambiguous. Counts before review are in Table 1.

Ambiguous files, 106 images in total, were then labelled by a single non-blinded rater, the author's assistant, who saw the prompt and the image. The assistant's labels were design 43, on-skin 3, incidental 10 and other 50. They are not a second independent annotation, and they were not blinded to the study aim. After that replacement, tattoo-subject images are those labelled design or on-skin. Preference probes and all four reward models on the reviewed HPDv2 tattoo-subject subset use the same 528 pairs over 125 prompts. ImageRewardDB uses 148 pairs over 9 prompts. The Pick-a-Pic tattoo-subject slice that remains after the CLIP filter is 65 pairs over 11 prompts. That slice is very small. ImageRewardDB tattoo-subject has only 9 prompts. Its intervals are wide because the bootstrap resamples prompts, not pairs.

4.3 Defect corpus

Controlled defects and the spread test use 120 designs drawn with seed 0 from Drozdik/tattoo_v0. That set is released for research under CC BY-NC-SA. We do not redistribute those images. Six operators at two levels produce 1,560 images including the clean copies. Reward scoring of defects uses 30 of the designs.

5 Results

5.1 Preference agreement

Table 2 is the primary result. Each cell is pairwise accuracy A, the effect $A - 0.5$, Cohen's h, and a Holm-adjusted prompt-cluster p. The bootstrap resamples prompts, not pairs. Figure 1

plots the same point estimates of A. The naive composite is not in this table. On HPDv2 tattoo-subject pairs, seven pre-registered probes have bootstrap intervals entirely below one half, and those seven remain significant after Holm ($p_{\text{Holm}} \leq 0.006$). The tight-gap fraction agrees on 0.335 of pairs (effect -0.165, $h = -0.34$, 0.288 to 0.383, n=528). Edge density is 0.335, specks 0.340, mid-tone fraction 0.369, border variation 0.377, sub-needle fraction 0.411, and minimum line width 0.428. Open ends are near chance at 0.528 (0.470 to 0.586). Post-hoc edge transition is 0.460 and was not a registered test. Annotators preferred the image the probes call harder to tattoo.

HPSv2.1 matches the human winner on 0.790 (effect +0.290, $h = +0.62$, $p_{\text{Holm}} \leq 0.006$, 0.751 to 0.827, n=528). PickScore matches on 0.723, CLIPScore on 0.682, and ImageReward on 0.684. HPSv2.1 was trained on HPDv2, so that row is in-distribution. It shows the scorer still ranks these pairs as the annotators did. It does not show a needle model.

On ImageRewardDB the probes sit near one half, on 9 prompts, and none of the 13 scorers is significant after Holm (0). Tight-gap accuracy is 0.500 (effect +0.000, $h = +0.00$, 0.390 to 0.606, n=148, p_{Holm}=1.000). Mid-tone fraction is the highest probe at 0.635. Edge density is the lowest at 0.378. ImageReward scores 0.615 (0.438 to 0.757) and was trained on this dataset. HPSv2.1 is 0.493 (0.363 to 0.644), PickScore 0.547, CLIPScore 0.554. These intervals are too wide to rank models.

On the Pick-a-Pic tattoo-subject slice the pre-registered probes do not repeat the HPDv2 pattern as a stable ranking. Tight-gap accuracy is 0.492 (0.361 to 0.717, n=65), edge density 0.400, specks 0.385, mid-tone fraction 0.585, and minimum line width 0.492. The intervals cover both sides of one half because there are only 11 prompts, and no probe is Holm-significant. The one Holm-significant scorer on this slice is PickScore (1 of 13, $p_{\text{Holm}} \leq 0.006$). HPSv2.1 matches on 0.754 (0.475 to 0.889) and ImageReward on 0.785 (0.550 to 0.900). Neither model was trained on Pick-a-Pic. PickScore matches on 0.815 and was trained on Pick-a-Pic v1, so that cell is leaky. CLIPScore is 0.723.

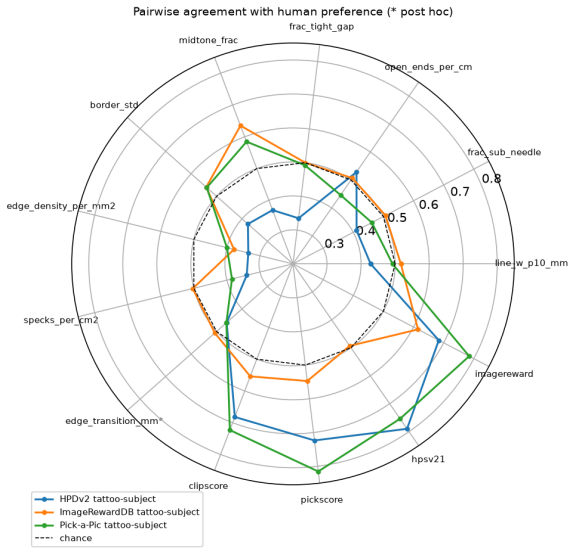


Figure 1: Per-dimension agreement with human preference on the three tattoo-subject slices. Each spoke is one scorer. This is the same comparison as Table 2, without the deprecated composite.

5.2 Fusion learns the opposite signs

A prompt-held-out logistic fusion of the eight pre-registered probes reaches 0.708 on HPDv2 (0.667 to 0.750, n=528). Adding edge transition leaves 0.701. The full-sample coefficients invert

Table 2: Pairwise accuracy A against human preference on tattoo-subject pairs, the effect $A-0.5$, and Cohen's h against 0.5. The resampling unit is the prompt (cluster bootstrap, 2000 resamples, seed 0). p_{Holm} is a two-sided bootstrap p after Holm correction within each dataset across all 13 scorers (the same count on each set: 13 and 13). A printed floor means $p_{\text{Holm}} \leq$ that value, which is the bootstrap resolution. Holm-significant scorers: 11/13 on HPDv2, 0/13 on ImageRewardDB, 1/13 on Pick-a-Pic. Pair counts are 528, 148 and 65. Edge transition is post hoc. HPSv2.1←HPDv2, ImageReward←ImageRewardDB and PickScore←Pick-a-Pic v1 are in-distribution.

	HPDv2				ImageRewardDB				Pick-a-Pic			
Scorer	A	$A-0.5$	h	p_{Holm}	A	$A-0.5$	h	p_{Holm}	A	$A-0.5$	h	p_{Holm}
line_w p10	0.428	-0.072	-0.14	0.006	0.517	+0.017	+0.03	1.000	0.492	-0.008	-0.02	1.000
frac sub-needle	0.411	-0.089	-0.18	0.006	0.507	+0.007	+0.01	1.000	0.462	-0.038	-0.08	1.000
open ends / cm	0.528	+0.028	+0.06	0.397	0.507	+0.007	+0.01	1.000	0.446	-0.054	-0.11	1.000
frac tight gap	0.335	-0.165	-0.34	0.006	0.500	+0.000	+0.00	1.000	0.492	-0.008	-0.02	1.000
mid-tone frac.	0.369	-0.131	-0.26	0.006	0.635	+0.135	+0.27	0.462	0.585	+0.085	+0.17	1.000
border std	0.377	-0.123	-0.25	0.006	0.541	+0.041	+0.08	1.000	0.538	+0.038	+0.08	1.000
edge density	0.335	-0.165	-0.34	0.006	0.378	-0.122	-0.25	1.000	0.400	-0.100	-0.20	1.000
specks / cm^2	0.340	-0.160	-0.33	0.006	0.503	+0.003	+0.01	1.000	0.385	-0.115	-0.23	1.000
edge transition	0.460	-0.040	-0.08	0.279	0.507	+0.007	+0.01	1.000	0.462	-0.038	-0.08	1.000
CLIPScore	0.682	+0.182	+0.37	0.006	0.554	+0.054	+0.11	1.000	0.723	+0.223	+0.46	0.955
PickScore	0.723	+0.223	+0.46	0.006	0.547	+0.047	+0.09	1.000	0.815	+0.315	+0.68	0.006
HPSv2.1	0.790	+0.290	+0.62	0.006	0.493	-0.007	-0.01	1.000	0.754	+0.254	+0.53	0.765
ImageReward	0.684	+0.184	+0.38	0.006	0.615	+0.115	+0.23	1.000	0.785	+0.285	+0.61	0.342

Table 3: Prompt-held-out logistic fusion (GroupKFold on prompt, k=5, seed 0). Accuracy is the fraction of pairs whose held-out predicted winner matches the human winner. Brackets are 95% prompt-cluster bootstrap intervals (2000 resamples). Coefficient signs are discussed in the text; magnitudes stay in `fusion_lopo.csv`. On Pick-a-Pic the reward fusion is led by PickScore, which was trained on Pick-a-Pic v1.

Model	HPDv2 acc. [CI]	n	ImageRewardDB acc. [CI]	n	Pick-a-Pic acc. [CI]	n
Probes, pre-registered	0.708 [0.667, 0.750]	528	0.500 [0.329, 0.702]	148	0.400 [0.266, 0.757]	65
Probes, including edge transition	0.701 [0.656, 0.745]	528	0.480 [0.308, 0.695]	148	0.385 [0.252, 0.711]	65
Rewards	0.790 [0.748, 0.827]	528	0.662 [0.545, 0.767]	148	0.831 [0.687, 0.939]	65
Rewards + all probes	0.767 [0.729, 0.803]	528	0.561 [0.403, 0.724]	148	0.785 [0.700, 0.896]	65

the a-priori signs for line width, sub-needle fraction, tight gaps, mid-tones, border variation, edge density and specks. Only open ends stay negative. The fusion predicts annotator taste by rewarding features the probes call worse to tattoo. Reward fusion reaches 0.790 (n=528) and is not improved by adding probes (0.767). In the reward-only fit the HPSv2.1 coefficient is positive and dominates; PickScore's coefficient is negative.

On ImageRewardDB the pre-registered probe fusion is 0.500 (0.329 to 0.702). Reward fusion is 0.662; adding probes lowers it to 0.561. The reward-only fit puts a positive sign on ImageReward and a negative sign on HPSv2.1. We do not treat the probe signs on 9 prompts as stable.

On Pick-a-Pic the pre-registered probe fusion is 0.400 (0.266 to 0.757, n=65), below one half, and the all-probe fusion is 0.385. The full-sample signs are not the HPDv2 inversion. Line width is negative, and edge density and specks are positive, so those three disagree with the a-priori tattooability signs. Sub-needle fraction, open ends, tight gaps, mid-tones and border variation keep the a-priori sign. Reward fusion reaches 0.831 (0.687 to 0.939); adding probes lowers it to 0.785. In the reward-only fit PickScore and ImageReward are positive and HPSv2.1 is negative. The PickScore term is the leaky one.

5.3 Injected defects

Table 4 asks whether a known damage lowers the score. At 5 cm, edge transition rates blur at 0.97 and 0.97, detail loss at 1.00 and 0.97, and thickening at 0.90 and 1.00. Mid-tone fraction

Table 4: Controlled-defect detection at a 5 cm print size: fraction of designs on which the clean image scores better than the degraded image (ties count as one half). Probe columns use a-priori signs. Reward columns use the raw reward (higher is better). The deprecated composite is not a column here; Appendix A states it in one sentence. n=30 designs per cell (the reward subset).

Defect	Lv.	trans.	sub-ndl	gap	mid	HPS	IR	Pick	CLIP
blur	1	0.97	0.00	0.10	0.93	0.80	0.90	0.53	0.97
blur	2	0.97	0.00	0.00	0.93	0.97	0.90	0.87	0.70
detail loss	1	1.00	0.13	0.20	0.97	0.97	0.87	0.83	0.77
detail loss	2	0.97	0.13	0.03	0.93	1.00	0.93	0.97	0.87
haze	1	0.47	0.83	0.97	1.00	1.00	0.77	0.80	0.57
haze	2	0.23	0.97	0.93	1.00	1.00	0.83	0.97	0.53
speckle	1	0.73	1.00	0.33	1.00	1.00	0.60	0.70	0.20
speckle	2	0.73	1.00	0.70	1.00	1.00	0.47	0.83	0.33
thicken	1	0.90	0.10	0.37	0.43	0.87	0.77	0.97	0.70
thicken	2	1.00	0.03	0.37	0.33	0.97	0.87	0.97	0.63
thin	1	0.83	0.50	0.07	0.33	0.90	0.83	0.97	0.47
thin	2	0.83	0.43	0.07	0.20	1.00	0.90	1.00	0.70

rates haze at 1.00 and speckle at 1.00. Sub-needle fraction rates thinning at 0.50 and 0.43. The deprecated composite is omitted here and stated once in Appendix A.

HPSv2.1's lowest cell in the table is light blur at 0.80. PickScore on that cell is 0.53. CLIP-Score on speckle is 0.20 and 0.33. ImageReward on heavy speckle is 0.47. Rewards detect many defects and do not name the millimetre quantity that moved. Figure 2 shows the grid.

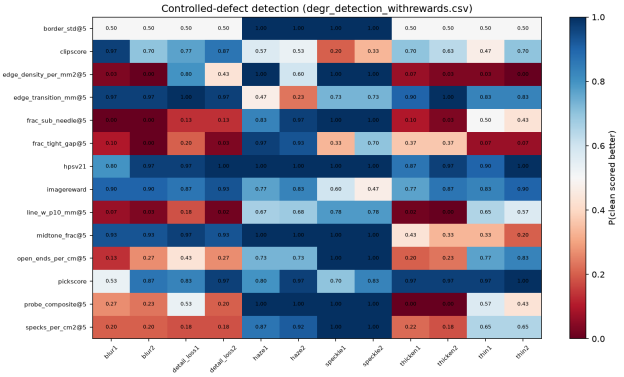


Figure 2: Detection rates on six injected defects at two strengths. The 5 cm numbers for the probes we interpret, and for the four rewards, are in Table 4.

5.4 Scale and spread

With no additive pixel noise, skeleton F_1 at spread σ=0 is 1.000 at s=3 cm, 1.000 at s=5 cm, 1.000 at s=10 cm and 1.000 at s=15 cm (Table 5, Figure 3). The same four sizes under the old σ=4 grey-level noise are 0.986, 0.972, 0.956 and 0.932, which fall as the print grows. That fall is not in the default grid. At 0.30 mm spread, default F_1 is 0.477, 0.563, 0.668 and 0.733. At 0.60 mm it is 0.238, 0.280, 0.391 and 0.479. Gap survival at 0.30 mm is 0.895, 0.918, 0.948 and 0.966. Table 5 reports a subset of that 0.05 mm grid.

Table 6 is the size at which mean skeleton F_1 first reaches seven tenths, and eight tenths. At zero spread both levels are met at 3 cm. At 0.20 mm the seven-tenths size is 5 cm and the eight-tenths size is 15 cm. At 0.30 mm the seven-tenths size is 15 cm and eight tenths is n/a. At

Table 5: Mean skeleton F_1 (tolerance 0.2 mm) and, on the following row, mean gap survival. Default simulation: no additive pixel noise, and the same Otsu threshold on the reference and the blurred image. Rows are longest-side print sizes in centimetres. Columns are a subset of the 0 to 0.60 mm grid (step 0.05 mm); the figure shows every step. Each cell averages 120 designs. At spread 0, F_1 is exact on this grid.

Size (cm)	0	0.15	0.30	0.45	0.60
3	1.000	0.757	0.477	0.328	0.238
3 (gap)	1.000	0.928	0.895	0.879	0.869
5	1.000	0.810	0.563	0.384	0.280
5 (gap)	1.000	0.951	0.918	0.899	0.888
10	1.000	0.850	0.668	0.505	0.391
10 (gap)	1.000	0.976	0.948	0.929	0.916
15	1.000	0.870	0.733	0.594	0.479
15 (gap)	1.000	0.987	0.966	0.948	0.935

0.60 mm neither level is reached (n/a, n/a).

At 5 cm and 0.30 mm, tight-gap fraction correlates with gap survival at -0.824 (partly circular), edge density with gap survival at -0.599, edge transition with skeleton F_1 at -0.247, and minimum line width with skeleton F_1 at 0.507. These associations live inside the simulator.

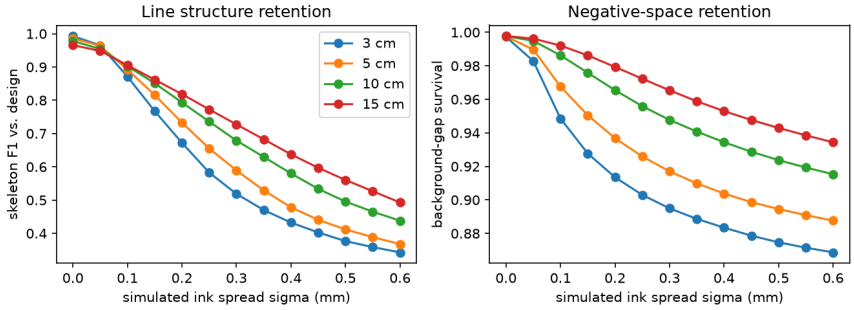


Figure 3: Mean skeleton F_1 against ink spread σ at each print size s. The grid is the default simulator: no additive pixel noise. Table 5 lists a subset of the same cells.

6 Discussion

The preference result and the defect result answer different questions, and they should not be averaged into one leaderboard.

On HPDv2 tattoo-subject pairs, human winners are the images our a-priori probes call worse to tattoo. Dense shading, tight spacing and thin lines are attractive on a screen. They are also the features a needle of width w_n and a gap of width g penalise. A prompt-held-out fusion can predict the annotator, at 0.708, only by flipping those penalties. Using that fusion as a tattooability reward would optimise the opposite of the probe specification. Using HPSv2.1 as a tattooability reward would do the same thing for a different reason: it agrees with the annotators

Table 6: Smallest tested print size, in centimetres, whose mean skeleton F_1 reaches seven tenths or eight tenths at that spread. `n/a` means no tested size up to 15 cm reached the level. Noise is off. This is a sensitivity table for the simulator, not a healed-skin recommendation.

Spread (mm)	Min. size, F_1 seven tenths	Min. size, F_1 eight tenths
0	3	3
0.10	3	3
0.20	5	15
0.30	15	n/a
0.40	n/a	n/a
0.50	n/a	n/a
0.60	n/a	n/a

(0.790) on data it was trained on, and it has no millimetre scale.

Edge transition separates blur, detail loss and thickening from the clean image. Mid-tone fraction separates haze and speckle. Sub-needle fraction separates thinning. The deprecated composite cancels some of those effects because thickening improves a width probe while hurting a gap probe. Per-dimension reporting is the property the reward models do not have: HPSv2.1 detects most injected defects and does not name which geometric fact changed.

The Pick-a-Pic tattoo-subject slice is only 65 pairs. Probe fusion there stays below one half (0.400) and does not reproduce the HPDv2 sign flip in full. Reward fusion is high (0.831) with a positive PickScore coefficient, which is training leakage from Pick-a-Pic v1, not a new external test. HPSv2.1 and ImageReward were not trained on that set; their Pick-a-Pic intervals are still wide.

ImageRewardDB does not rescue a positive probe–preference correlation. The point estimates sit near chance, the prompt count is 9, and ImageReward's own accuracy on that set is an in-distribution number with a wide interval. A claim that "rewards ignore tattooability" is too broad: several rewards notice blur, haze and thinning in the injection study. The supported claim is narrower. Preference on the public tattoo-subject pairs we could score is not a tattooability label, and models trained to imitate that preference should not be described as if it were.

A practical screen for AI tattoo generation can reject sub-needle line mass, tight gaps and soft transitions at the intended print size s. Those gates disagree with current preference rewards on the HPDv2 pairs above. That disagreement is a reason to keep the gates.

7 Limitations

Tattoo-subject prompt counts are small outside HPDv2. ImageRewardDB contributes 9 prompts and Pick-a-Pic 11. Cluster-bootstrap intervals on those sets are wide, and after Holm no probe is significant (0 of 13 scorers on ImageRewardDB; the single Pick-a-Pic rejection is PickScore, 1 of 13). Those rows do not support a model ranking.

The class filter mixes keywords, CLIP zero-shot, and a single non-blinded annotator on 106 ambiguous images. A design-versus-portrait error changes which pairs enter Table 2.

Needle width w_n and gap width g are practitioner-scale assumptions, not measurements from this study. Other machines, other skin, and colour packing change both numbers. The ink-spread simulator is an isotropic Gaussian at a stated σ. It is unvalidated on healed skin. Real ink does not spread as a circular blur, and we have no paired photograph of a design before and after healing. With noise off, skeleton F_1 at σ=0 is 1.000 at every tested size s, including 15 cm (1.000). The earlier decline with size was grey-level noise of pixel standard deviation 4, which leaves 0.986 at s=3 cm and 0.932 at s=15 cm. Removing that artifact does not make the Gaussian a model of healed skin. At 0.60 mm spread, no tested size reaches mean F_1 of seven tenths (n/a).

Reward-model scores leak their training sets. HPSv2.1 was trained on HPDv2, ImageReward

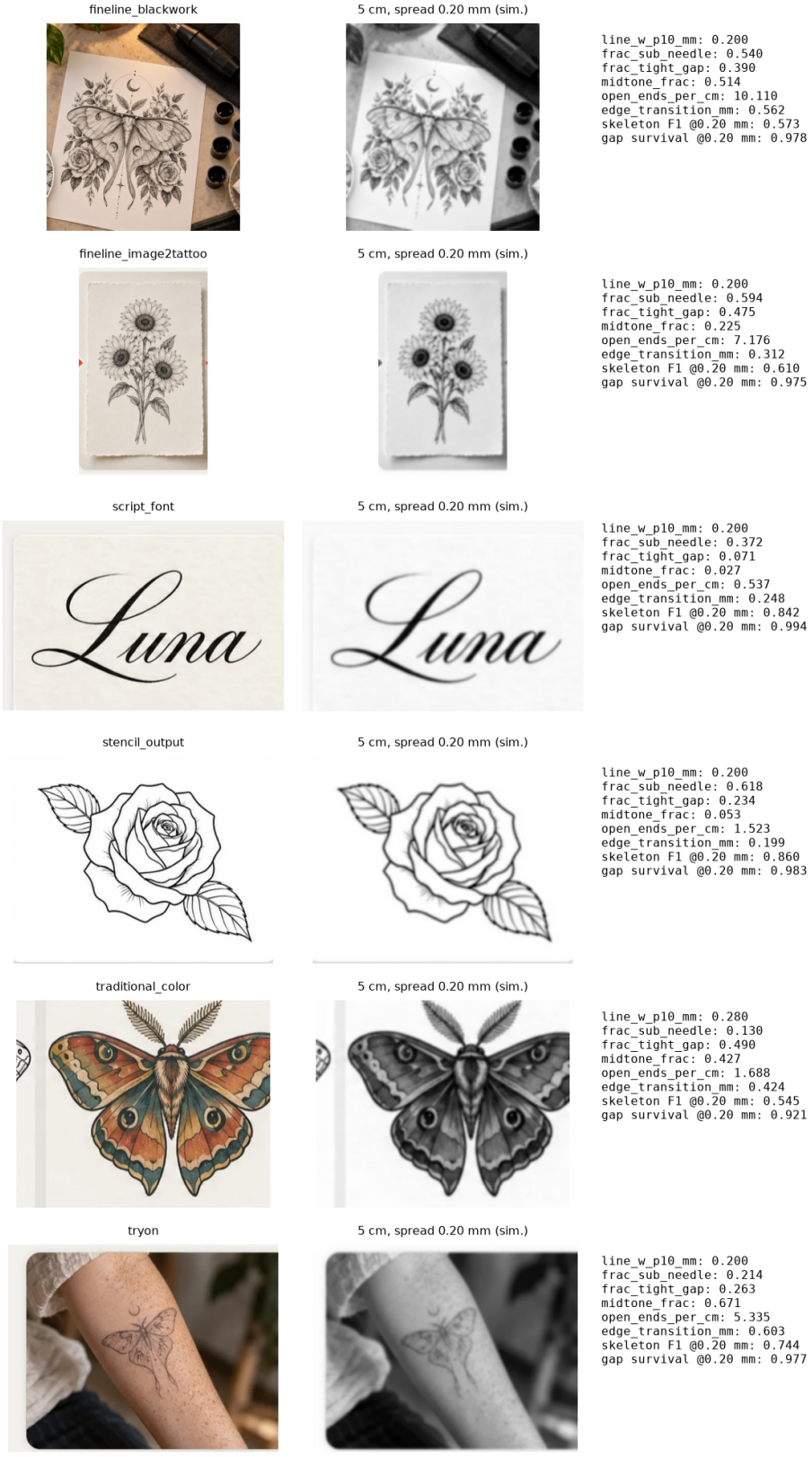


Figure 4: The author operates tatany.app. The images are tatany.app product case images (marketing mock-ups cropped to one panel; provenance in supplementary notes). Probes at s=5 cm and a simulated spread σ=0.20 mm are shown for illustration only. tatany is not ranked, evaluated or compared with any other system.

on ImageRewardDB, and PickScore on Pick-a-Pic. We cannot remove the tattoo-subject pairs from those training sets retroactively. CLIPScore was not trained as a preference model. The Pick-a-Pic licence is uncertain: the original dataset card has been removed, and mirrors say MIT. We did not verify that statement.

Otsu thresholding and the greyscale conversion assume a dark-ink-on-light-paper drawing. Colour work, skin photographs and mid-grey packing violate that assumption, and the probes still emit a number. Edge-transition width is a post-hoc probe, chosen after the first controlled-degradation run. Its preference accuracy is exploratory. Defects are synthetic and one at a time, and reward detection uses 30 designs. The composite mixes probes whose signs fight each other under thickening. It is deprecated and is not a column of Table 4.

Figure 4 is a marketing mock-up. The panels are tatany.app product case images, cropped to one panel, not logged generations. Background paper and skin texture change what a probe would read, which is why the figure is illustration only and why per-image probe values are not quoted. No commercial AI tattoo generator was ranked. Generalising from HPDv2 to images produced by a product model would be a new study.

8 Conclusion

Preference is not tattooability on the pairs we audited. On HPDv2 tattoo-subject pairs, pre-registered probes that penalise dense, shaded, thin-line images agree with human winners below chance. The tight-gap probe, for example, agrees on 0.335 of pairs. A prompt-held-out fusion of the probes reaches 0.708 by assigning the opposite signs. HPSv2.1 agrees with those annotators at 0.790 on data it was trained on. ImageReward agrees with ImageRewardDB annotators at 0.615, with a wide interval and only 9 tattoo-subject prompts.

The same probes, read one at a time, detect the injected defects they were defined for, and the reward models detect many of those defects too, without a physical unit. A Pick-a-Pic slice of 65 pairs does not overturn the HPDv2 result, and PickScore on it is in-distribution. A Gaussian scale-and-spread stress test, with no additive pixel noise, keeps skeleton F_1 at 1.000 when the spread is σ=0 and lowers it as σ grows. It remains an assumption, not a skin study.

For AI tattoo generation, a preference reward is evidence about annotators. A millimetre probe is evidence about a declared needle, gap and print size. Reporting those probes separately, and stating when a reward model is scored in distribution, is the audit.

Resources and data availability

Probe definitions, degradation operators and the analysis scripts live with the study code that produced `numbers.tex`. HPDv2, ImageRewardDB and Pick-a-Pic are used under the terms of their public releases [3–5]. Drozdik/tattoo_v0 is CC BY-NC-SA and was used for research only; we do not redistribute it. Pick-a-Pic metadata and images were read locally from a mirror whose MIT statement we could not verify against the removed original card. They are not redistributed. Derived pair counts and scores are the tables in this paper.

Declaration of competing interest

The author operates tatany.app, a commercial AI tattoo generator. That conflict is relevant because a published critique of generic preference rewards could favour a product that advertises tattoo-specific checks. tatany was not evaluated or ranked. No tatany output appears in HPDv2, ImageRewardDB, Pick-a-Pic, the defect set, or the spread test. The probes were not tuned on tatany images.

Acknowledgements

We thank the maintainers of HPDv2, ImageRewardDB, Pick-a-Pic and Drozdik/tattoo_v0 for releasing data. The ambiguous-class review was done by the author's assistant, as a single non-blinded rater.

A Probe implementation notes

Distance transforms use a 5×5 mask. Skeleton endpoints are pixels whose 3×3 neighbour sum equals 2, and the open-end rate divides that count by skeleton length in centimetres. Tight gaps use the background inside the ink box only. Specks are 8-connected ink components smaller than $(0.30 \times 10)^2$ pixels. Mid-tones lie strictly inside grey $(60, 200)$, the border band is 5 px, and Canny hysteresis is 50 and 150 [26]. Otsu is replaced by a fixed mid-grey cut when the grey standard deviation is at most 1 [25]. Thinning is Zhang–Suen [27]. Edge transition divides median paper-minus-ink contrast by median Sobel magnitude on Canny pixels (3×3 Sobel, one-eighth scale) and divides by 10 px/mm. Empty masks yield 0. The deprecated composite is the mean of signed robust z-scores, using the median and interquartile range inside the scored pool. On the defect subset it rates haze and speckle at 1.00 and 1.00, thickening at 0.00 and 0.00, and light blur at 0.27. Thickening moves the width term and the gap term in opposite directions, which is why that average is not a primary result.

The spread test blurs with σ in millimetres times 10 px/mm and, in the reported grid, adds no pixel noise. Skeleton F_1 uses a square tolerance of width $2 \times \mathrm{round}(0.2 \times 10) + 1$ pixels. The reference mask and the spread mask use one shared Otsu threshold. SSIM is not computed [28]. Bootstrap, fusion, design sampling, speckle and haze noise use seed 0.

B Pre-registration record

Phase-1 probes and signs, fixed before any preference accuracy was computed, are the eight quantities in Section 3: line width +1; sub-needle fraction, open ends, tight gaps, mid-tone fraction, border variation, edge density and specks all −1. The primary estimand is pairwise accuracy on tattoo-subject pairs, with a 2000-resample prompt-cluster bootstrap and seed 0, at a 5 cm print size, needle 0.30 mm and gap 0.50 mm. The unweighted signed composite was defined in advance. It is in Table 4 and is deprecated in favour of Table 2.

`edge_transition_mm` (sign −1) was added after the first controlled-degradation run, which showed that blur and detail loss did not reliably shrink lines below 0.30 mm or create specks. Its detection rates are descriptive. Its preference accuracies, 0.460 on HPDv2, 0.507 on ImageRewardDB and 0.462 on Pick-a-Pic, were not registered tests. The pre-registered fusion excludes it. The fusion that includes it does not change the conclusion. Full-sample HPDv2 probe coefficients invert the registered signs for every pre-registered feature except open ends, which stay negative. Those inverted signs are part of the negative result, not a specification we chose in advance.

Automatic labels left 99 HPDv2 images and 7 ImageRewardDB images ambiguous, 106 images in total. A single non-blinded rater, the author's assistant, assigned design (43), on-skin (3), incidental (10) and other (50). The rater saw the prompt and knew the labels would define the tattoo-subject subset. This was not a blinded adjudication. Counts after replacement are the "after review" rows of Table 1.

Pick-a-Pic scoring was added after the HPDv2 and ImageRewardDB runs. Keyword selection yielded 286 labelled pairs over 23 prompts; after CLIP filtering, 65 design or on-skin pairs over 11 prompts remain. That analysis was not in the Phase-1 registration. PickScore's accuracy on it is leaky because PickScore was trained on Pick-a-Pic v1. The Gaussian grid is a sensitivity check, not a registered test on healed skin. Its default has no additive pixel noise, so skeleton F_1

Table 7: Spearman rank correlation, at 5 cm and 0.30 mm spread, between a clean-image probe and the simulated outcome on that design (n=120). Gap survival versus tight-gap fraction is partly circular: both read narrow background gaps. These correlations are not a healed-skin validation.

Probe on the clean design	ρ vs. skeleton F_1	ρ vs. gap survival
line_w p10	0.507	0.288
frac sub-needle	-0.482	-0.111
open ends	-0.075	0.042
frac tight gap	-0.476	-0.824
mid-tone frac.	0.006	-0.462
border std	0.145	0.146
edge transition	-0.247	-0.339
edge density	-0.298	-0.599
specks	-0.343	-0.264
ink fraction	-0.116	-0.483

at spread 0 is 1.000 at 5 cm rather than the noise-on value 0.972. Reward detection of injected defects uses 30 designs after a probe pass on 120 designs, and is not a registered superiority test. Training overlap for HPSv2.1 and ImageReward was known from the dataset papers before scoring [3, 4]. A title-and-abstract search did not find a prior audit of preference rewards against millimetre tattoo probes. Semantic Scholar and Google Scholar were not fully queried.

Spearman correlations at 5 cm and 0.30 mm spread, each over 120 designs, are in Table 7. The headline four are also stated in Section 5.

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